AMPLENESS OF A LINE BUNDLE ON ONE FIBRE OF A PROPER MORPHISM SPREADS TO AN OPEN NEIGHBOURHOOD OF THE BASE POINT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 14153-14163 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 14165-14194, the complete author proof (28 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The proof closes with author-supplied hints rather than a line-by-line verification of the last pullback; the mechanism (affine base change preserving ampleness) is fully named in the span. Reference block cites \cite[IV Corollary 9.6.4]{EGA}; bibliography.tex with the snapshot my.bib EGA entry attached.

Editorial note. The statement's reference environment cites \cite[IV Corollary 9.6.4]{EGA}; a bibliography.tex holding the snapshot my.bib EGA entry is attached (precedent L16058/L16018).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The Noetherian approximation of proper maps (Limits, proposition and lemma-proper-limit-*), descent of invertible modules to a finite stage (Limits, lemma-descend-invertible-modules), the limit-detection of ampleness (Limits, lemma-limit-ample), and the finite-type openness of ampleness (Cohomology of Schemes, lemma-ample-in-neighbourhood).

Source reference. Original lemma 0D2N: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14194–14194. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B8W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14194–14194. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09MT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14194–14194. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0A0P: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14194–14194. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0893: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14194–14194. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0892: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14194–14194. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. [1, IV Corollary 9.6.4] *Let $f : X \to Y$ be a proper morphism of schemes. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $y \in Y$ be a point such that $\mathcal{L}_y$ is ample on X_y. Then there is an open neighbourhood $V \subset Y$ of y such that $\mathcal{L}|_{f^{-1}(V)}$ is ample on $f^{-1}(V)/V$.*

Proof. We may assume Y is affine. Then we find a directed set I and an inverse system of morphisms $X_i \to Y_i$ of schemes with Y_i of finite type over $\mathbf{Z}$, with affine transition morphisms $X_i \to X_{i'}$ and $Y_i \to Y_{i'}$, with $X_i \to Y_i$ proper, such that $X \to Y = \lim(X_i \to Y_i)$. See Limits, Lemma 0A0P. After shrinking I we can assume we have a compatible system of invertible $\mathcal{O}_{X_i}$-modules $\mathcal{L}_i$ pulling back to $\mathcal{L}$, see Limits, Lemma 0B8W. Let $y_i \in Y_i$ be the image of y. Then $\kappa(y) = \operatorname{colim} \kappa(y_i)$. Hence for some i we have $\mathcal{L}_{i,y_i}$ is ample on X_{i,y_i} by Limits, Lemma 09MT. By Cohomology of Schemes, Lemma 0D2N we find an open neighbourhood $V_i \subset Y_i$ of y_i such that $\mathcal{L}_i$ restricted to $f_i^{-1}(V_i)$ is ample relative to V_i. Letting $V \subset Y$ be the inverse image of V_i finishes the proof (hints: use Morphisms, Lemma 0893 and the fact that $X \to Y \times_{Y_i} X_i$ is affine and the fact that the pullback of an ample invertible sheaf by an affine morphism is ample by Morphisms, Lemma 0892). □

References

[1] Dieudonné, Jean and Grothendieck, Alexander. *Éléments de géométrie algébrique*. Inst. Hautes Études Sci. Publ. Math. 4, 8, 11, 17, 20, 24, 28, 32 (1961–1967).